Sahaj Bhadja

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Summary

Software Engineer pursuing an MSCS at USC with 2+ years building production systems and autonomous-driving research pipelines. Skilled in **C++**, Python, Linux, stateful workflows, integration testing, debugging, and reliable software delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Operating Systems, Software Engineering, Database Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Structures, Object-Oriented Programming, Computer Networks, Machine Learning

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Restructured fault-tolerant asynchronous workflows with explicit **state transitions**, failure isolation, and FastAPI workers, processing 1K+ concurrent jobs at triple throughput while sustaining 99%+ reliability across 5K+ monthly tasks.
- Diagnosed distributed data-flow bottlenecks across Azure SQL and cloud storage, reducing latency from 1.2 seconds to 350 milliseconds through **debugging**, indexed queries, validation checks, and reproducible release procedures.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Developed concurrent Python services for secure file ingestion and multimodal routing, processing 1K+ files per batch with fault isolation, provider fallbacks, and **interface validation** across production workflows.
- Built React and Flask interfaces supporting 150+ instructors and 10K+ daily requests, using REST APIs, automated checks, and **CI/CD release validation** to reduce production feedback turnaround 60%.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a Decision Transformer for **autonomous-driving behavior planning**, improving simulated trajectory planning 18% while processing 100K+ driving logs through reproducible Python evaluation pipelines and controlled baseline comparisons.
- Architected a spatial-temporal encoder combining HD-map geometry, lane context, and ego trajectories, improving closed-loop stability 15% through **simulation testing**, systematic failure analysis, baseline comparisons, and documented research evaluations.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Engineered cross-platform integration modules using **object-oriented design patterns**, improving API response times 30%, reducing technical debt 20%, and increasing maintainability across distributed Windows, Linux, and macOS deployments.
- Automated 20+ **unit and integration tests** with PyTest, Mockito, Jenkins, and virtual-machine builds across three operating systems, supporting root-cause debugging, Agile reviews, defect reporting, and reproducible release validation.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight modular services for authentication, resources, billing, leads, permissions, and notifications, synchronizing 2M+ distributed records through **real-time interfaces** while reducing staged production conflicts 33% during releases.
- Implemented automated unit, integration, and end-to-end tests through Jenkins CI/CD and Docker staging, achieving 95%+ **build validation** coverage and reducing deployment time 50% across controlled production release cycles.

Skills

Languages: C++, Python, Java, C#, SQL, TypeScript, JavaScript, Bash
ADAS & Systems: Behavior Planning, State Machines, Decision Making, Vehicle Dynamics, ROS 2, DDS, Linux, Embedded Systems, Multi-ECU Integration, Signal Mapping, Timing Analysis, Asynchronous Programming, Concurrency, OOP, Design Patterns
Testing: Unit Testing, Integration Testing, SIL, HIL, Test Benches, Simulation, GoogleTest, PyTest, Regression Testing, Test Automation, Debugging, Log Analysis, Requirements Traceability
Tools & Delivery: Git, CMake, Jira, Jenkins, GitHub Actions, CI/CD, Docker, REST APIs, FastAPI, AWS, Azure, PostgreSQL, SQL Server, Technical Documentation, Agile/Scrum, Release Management

Projects

UrbanDrive Behavior Planner | *C++, ROS 2, CARLA, DDS, GoogleTest, Linux*

- Developed a C++ **ADAS behavior-planning state machine** for urban driving, parking, and AEB scenarios, defining activation, arbitration, abort, and fault-handling logic with ROS 2 interfaces and simulation tests.

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2*

- Implemented FCFS, SJF, SRT, Round Robin, priority, and multilevel-queue **scheduling algorithms** in C++ with interactive visualization, validating 1K+ process combinations and reducing average turnaround time up to 25%.

Chat2SQL | *Java, TypeScript, Docker, Maven, PostgreSQL, SQL Server*

- Extended a cross-platform Java and TypeScript workspace supporting 10+ databases, using Docker and Maven to validate **interface compatibility**, schema synchronization, and reproducible deployments across Windows, macOS, and Linux.